

Managing Complex Campus Networks Using Secure Software Defined Networking (SDN)

Nikhil

Computer Science and Engineering

Chitkara University Institute of Engineering and Technology, Punjab, India

nikhil609.be22@chitkara.edu.in

Nampreet Singh

Computer Science and Engineering

*Chitkara University Institute of Engineering and
Technology, Punjab, India*

nampreet596.be22@chitkara.edu.in

Bharat

Computer Science and Engineering

*Chitkara University Institute of Engineering and
Technology, Punjab, India*

bharat212.be22@chitkara.edu.in

Jatin Sharma

Computer Science and Engineering

*Chitkara University Institute of Engineering and
Technology, Punjab, India*

jatin1714.be22@chitkara.edu.in

Abstract - The purpose of this paper is to explore the ways that Software Defined Networking (SDN), as a technology, can assist with the management and configuration issues faced by large, complex campus networks. Through the use of SDN, management of access to networks can be centralized, prioritization of types of traffic can be automated, and enforcement of security policies can be accomplished through centralizing control of networks as well as the execution of policy.

Moreover, the application of SDN enables network administrators to effectively respond to the dynamic requirements and changing conditions associated with network usage in a campus environment. In this paper we will explore the many advantages of implementing SDN in a campus network, and we will also address some of the challenges involved with deploying SDN solutions such as issues related to scaling and performance.

In summary, there are numerous examples of successful deployments of SDN solutions in higher education and in large college campus networks. By implementing the SDN architecture into your campus network you will experience improved operational management and a reduction in the complexity of traditional campus network management, leading to an overall improvement in network performance for both network administrators and end users. Ultimately, implementations of SDN will improve the total quality of experience for network administrators and users at all levels within the campus network.

Keywords— Network, Architecture

I. INTRODUCTION

In today's rapidly changing world of education with digital technologies changing every day, managing school networks is a very big challenge for schools. Existing methods to manage networks don't provide enough control and security due to the increasing number of digital devices and also the demand from users for fast connections with strong security measures. Fortunately, a cutting-edge technology called Secure Software Defined Networking (SDN) has been developed to solve these complex problems. Secure SDN combines the features of software defined networking (e.g. speed/ flexibility) with the secure features of a public key infrastructure. The combination of these two technologies revolutionizes how to manage school networks and provides an easy way to manage school networks. The introduction of secure SDN to manage school networks introduces a major change in the way schools design, implement, and manage their networks by separating the control plane from the data plane.[1]

Secure Software Development Network (SDN) is greatly improving the procedures and practices around the Design and Management of the network. Through a Centralized Management System and a Security Framework, SDN allows you to centrally manage the Configuration, Monitoring, & Security of your entire infrastructure from one console. The distinct architectural features of this implementation increase Customization and Service Management and provide security to ensure your sensitive data and critical network services maintain their integrity from potential compromises. The Combination of the previous mentioned security features make Secure SDN an essential resource for your campus, ensuring operations are Secure and Protected.

The innovative concept of Software Defined Networking (SDN) originated at Stanford University through the Clean Slate research initiative in 2006. In 2009, the de-coupling of control/data forwarding laid the groundwork for a new era of networking activities. Because there is now a central controller executing network management functions (e.g., configuration, RADIUS, NTP, load-balancing, etc.) while data is sent to switches, management performance has improved, and costs are reduced.

Google's successful use of SDN to build and manage its backbone in 2012 highlighted SDN's potential. This result has greatly increased the effective use of a large network (Google's backbone). The success of Google with SDN has also encouraged other large corporations around the world to create their own SDN solutions. Today, SDN has become a primary focus of many technology companies and will be a key part of the next generation of the Internet.

University networks are an internal telecommunications network for the campus. These networks provide a wide-ranging event and activity experience throughout the University in a very dynamic environment. The University manages the telecommunications configuration of interconnected LANs over a relatively small geographical area of the campus, and the University owns and manages all telecommunications hardware devices, such as switches and routers.

Network availability is critical for operational, instructional, and research activities in the current digital age of technology, therefore; having a reliable telecommunications network infrastructure is extremely important for any University. The University's telecommunications infrastructure must be secure, high-performing, and easily accessible by having features designed to provide remote access and centralized control. In many instances, building a campus-wide network can be complicated due to having to consider diverse manufacturers with many different product offerings for each application and telecommunication service.[2]

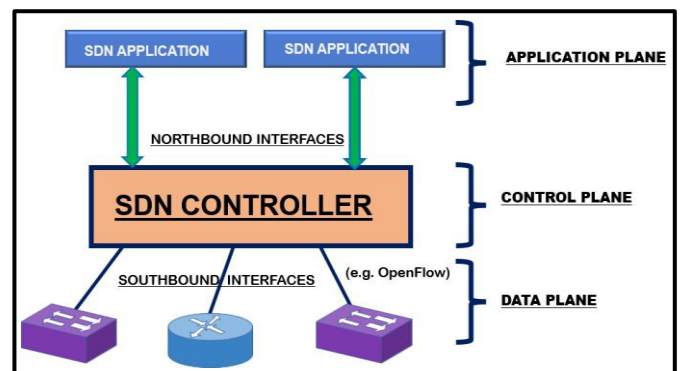


FIG. 1 SDN Controller

This paper outlines the three major areas of popular implementations of software-defined networking, detailing and discussing the roles and features of the Control Plane, the Data Plane, as well as integration. Current research in the area of Data Plane and Control Plane will also be discussed in detail, along with an exhaustive account of all proposed SDN implementations. By examining these items, we will begin to understand the effectiveness of SDN technology and how SDN technology is having an impact on the world we live in. In addition, this examination of these factors in combination with our understanding of the nuances of SDN technology and current research activity will enable us to maximize the potential benefits of SDN in the area of campus management and in general innovation in the industry.

In the future, SDN will continue to provide numerous applications and have significant ramifications across many industries. As a result of its reliability and flexibility, SDN can provide a multitude of benefits to telecommunications, data centres, cloud computing etc. SDN provides centralized management of networks, customizable network services, and state-of-the-art security capabilities, enabling unprecedented levels of performance, scalability, and efficiency. As SDN continues to develop, it will incorporate additional/new technologies (e.g., artificial intelligence).

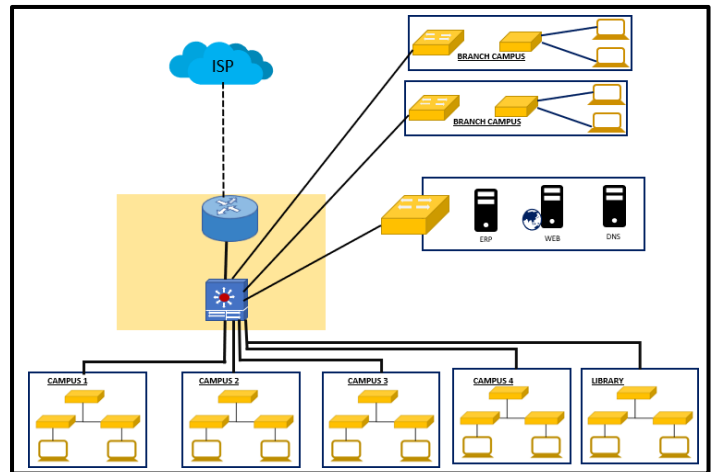


FIG. 2 Traditional Campus Network Design

II. LITERATURE SURVEY

In this research, literature reviews the use of Software-Defined Networking (SDN) as a means to improve the management of campus networks and how it can improve the way these networks are managed. Previous studies have looked at how SDN can scale; its effect on performance; and the many benefits of SDN, such as its ability to reduce configuration complexity, improve security and provide dynamic adaptation. This paper expands on previously published work by suggesting future research and development to optimize SDN within campus networks, thereby improving the efficiency with which campus networks are managed while improving overall user experience on those networks.

TABLE I [LITERATURE SURVEY]

Reference Number	Author(s) & Journal Details	Description/Interpretation
[3]	Zhao Zhang, Hailong Li, Siqi Dong, Lei Hu, 2018	Proposed Work: The proposed work analyses the development background, architecture and key technologies of Software-Defined Networking (SDN). It investigates the application fields of SDN and compare the advantages and disadvantages of different technologies. The research also explored cost control and security optimization in an SDN network. Future Work: Future work focuses on conducting researches and proposing innovative approaches to enhance cost control and security in SDN. Also, exploration of potential application directions for SDN will take place, considering emerging technologies and evolving network requirements.

[4]	Rajat Chaudhary, Anish Jindal, Gagangeet Singh Aujla, Shubhani Aggarwal, Neeraj Kumar, Kim-Kwang Raymond Choo, August 2019	<u>Proposed Work:</u> The proposed work aims to tackle the obstacles involved in crafting a secure energy exchange habitat for unmanned vehicles, particularly electric vehicles. BEST, a Blockchain-based secure energy trading mechanism, is put forth for this purpose. Software-defined networking facilitates low latency and real-time services, channelling EVs' petitions to a universal SDN controller. <u>Future Work:</u> The forthcoming undertakings will involve further development and optimization of the BEST scheme for secure energy trading. It will include enhancing the scalability, efficiency, and security of the blockchain-based validation process. Also, exploring the integration of emerging technologies such as edge computing and artificial intelligence for improved energy trading performance and decision-making in real-time scenarios.
[5]	Pradeep Kumar Sharma, S. S. Tyagi, November 2019	<u>Proposed Work:</u> Proposed system aims to advance network security by implementing an SDN-based application. Key areas for future development consider the integration of sophisticated security algorithms, exploration of distributed approach for the threat detection, utilizing machine learning methods, standardization of northbound API and development of specialized security applications, customized to meet the diverse network requirements. <u>Future Work:</u> One possibility for future work is to enhance the proposed SDN-based security application by integrating advanced security algorithms. Additionally, exploring distributed approaches for network threat detection and incorporating machine learning techniques can lead to a more efficient and effective dynamic response to evolving network threats.
[6]	Piyush Kumar Sharma, Devashish Gosain, Himanshu Sagar, Chaitanya Kumar, Aneesh Dogra, Vinayak Naik, H.B. Acharya, Sambuddho Chakravarty, 2020	<u>Proposed Work:</u> The proposed work focuses on addressing the implementation problems of Decoy Routing (DR) in censorship circumvention by introducing a novel system called Siege Breaker (SB). SB utilizes an SDN-based architecture and divides the tasks of DR into three independent modules: 1. an SDN, 2. a reconfigurable SDN switch 3. a secret proxy server <u>Future Work:</u> Future work will involve further development and optimization of the Siege Breaker system for efficient Decoy Routing. This includes refining the privacy-preserving signalling scheme and exploring additional techniques to enhance the performance and reliability of the proxy server. Also, conducting real-world deployments and evaluating the system's effectiveness and scalability in diverse network environments. The research considers potential integration with other emerging technologies, to improve the overall performance and user experience of Decoy Routing.
[7]	Faridullah Amarkhil, Prashansa Taneja, June 2020	<u>Proposed Work:</u> The proposed work constructs a security-enhanced control system for Wi-Fi devices, which is grounded on software-defined networking (SDN) in a wireless system. The objective is to enrich the management of networks, mobility of clients, and security in the SDN-WLAN infrastructure, explicitly in small business networks. Moreover, the endeavour tries to unite enterprise WLAN services and elevate the overall security parameters of SDN-WLAN for unparalleled network management and protection. <u>Future Work:</u> In the future, researchers will explore the potential of SDN-WLAN in substituting traditional networking tools and facilitating direct communication with diverse services. The unified portal enables the execution of new functions. Furthermore, network virtualization will be prominent to augment network speed and cope up with the progressive exigencies of IT network.

[8]	Samer Khamaiseh, Abdullah Al-Alaj, Mohammad Adnan, Hakam W. Alomari, May 2020	<u>Proposed Work:</u> The proposed work evaluates supervised and semi-supervised classifiers for detecting known and unknown DDoS saturation attacks in SDN. Different classifiers are tested on various datasets, and the results show that supervised classifiers perform well for known attacks, while semi-supervised classifiers excel in detecting unknown attacks. Additionally, an enhanced hybrid approach combining supervised and semi-supervised classifiers is proposed to improve detection of both known and unknown attacks. <u>Future Work:</u> In the future, the suggested methods will be expanded to multi-controller SDN networks to evaluate their performance in vast, complex environments. The emphasis will be on appraising the efficacy of supervised and semi-supervised classifiers in detecting both known and unknown attacks in such intricate network configurations. The aim is to further amplify the detection capabilities and scalability of the proposed techniques in practical SDN deployments with multiple controllers.
[9]	Junjie Zhang, Minghao Ye, Zehua Guo, Chen-Yu Yen, and H. Jonathan Chao, April 2020	<u>Proposed Work:</u> The proposed work focuses on developing CFR-RL, a reinforcement learning-based scheme for critical flow rerouting in SDN networks to balance link utilization and mitigate network disturbance caused by frequent rerouting. We formulate and solve a simple LP problem to reroute selected critical flows, achieving near-optimal performance by rerouting only a limited portion of total traffic. <u>Future Work:</u> In the future, the plan is to explore alternative objectives for CFR-RL, such as minimizing overall end-to-end delay in the network, and investigate the trade-off between maximizing performance and minimizing rerouting traffic. Also, the aim is to address scalability challenges by exploring multi-controller architectures and implementing retraining techniques to adapt CFR-RL to dynamic network conditions.
[10]	Hnin Thiri Zaw1, Aung Htein Maw, August 2019	<u>Proposed Work:</u> The proposed work focuses on designing a new architecture for dynamic multipath-based traffic management in Software-Defined Networks (SDN). The architecture includes components for detecting long flows, computing shortest paths, estimating delay and bandwidth utilization, calculating least cost paths, and rerouting traffic. Through simulations using Mininet and ONOS controller, it is demonstrated that this approach outperforms ECMP and reactive forwarding methods for both TCP and UDP traffic. <u>Future Work:</u> In the future, it is planned to further investigate and evaluate the proposed traffic management scheme in a realistic SDN testbed, moving beyond the Mininet emulator. It is also planned to explore the scalability and performance of the approach in larger networks and different network topologies.
[11]	Md. Jahidul Islam, Anichur Rahman, Sumaiya Kabir, Md. Razaul Karim, Uzzal Kumar Acharjee, Mostofa Kamal Nasir, Shahab S. Band, Mehdi Sookhak, Shaoen Wu, March 2021	<u>Proposed Work:</u> The authors present a distributed and decentralized Blockchain-SDN architecture for IoT in smart cities, focusing on scalability, security, and energy efficiency. They introduce a Cluster Head Selection algorithm and evaluate the performance using various parameters, demonstrating higher throughput, lower response time, and reduced gas consumption compared to existing works. <u>Future Work:</u> In their future research, the authors aim to develop an energy-efficient edge computing model by grasping SDN, NFV, and Blockchain technologies together. They will also cater to additional factors like end-to-end delay, network bandwidth, and various passive and active attacks. By considering these ongoing challenges, they aspire to further augment the performance and capabilities of the proposed architecture.

[12]	Ashutosh Kumar Singh, Naveen Kumar, Shashank Srivastava, February 2019	<u>Proposed Work:</u> This study focuses on efficiently deploying multiple controllers in software-defined networks (SDN) to minimize latency. The authors use optimization algorithms (TLBO and PSO) to address the controller placement problem (CPP) and improve the performance and reliability of SDN networks. The experimental results highlight the effectiveness of TLBO over PSO for publicly available topologies. Future research aims to explore alternative optimization algorithms for solving the CPP. <u>Future Work:</u> Further research will be conducted to investigate alternative optimization algorithms and propose new approaches for solving the controller placement problem (CPP) in software-defined networks (SDN). The aim is to enhance the efficiency and performance of SDN networks beyond the existing TLBO and PSO algorithms.
[13]	Tiago Linhares, Ahmed Patel, Ana Luiza Barros, Marcial Fernandez, 2022	<u>Proposed Work:</u> The proposed work aims to mitigate DDoS attacks in SDN networks by leveraging statistical analysis and traffic entropy. It introduces a mechanism to detect anomalies in packet flows and block malicious packets, enhancing network security and resilience. A prototype implementation in the Floodlight controller and testing in the Mininet environment validate the effectiveness of the proposed solution. <u>Future Work:</u> In future research, it is suggested to explore approaches to further reduce the time for identifying DDoS attacks in SDN networks. Real-world or pilot deployments of the SDN Truth system in large-scale SDN networks would be valuable to evaluate its viability, scalability, and effectiveness.
[14]	A. Al-Jawad, I. -S. Comşa, P. Shah, O. Gemikonakli, R. Trestian, December 2021	<u>Proposed Work:</u> This paper introduces a Reinforcement Learning (RL)-based framework for multimedia-based SDN environments to improve QoS provisioning by selecting optimal routing algorithms. The proposed solution outperforms state-of-the-art approaches in terms of throughput, packet loss, and user experience. <u>Future Work:</u> Future research can focus on enhancing the RL-based framework using advanced reinforcement learning techniques. Additionally, scalability and real-world deployment should be explored to evaluate the framework's effectiveness in diverse network scenarios and traffic conditions.
[15]	Yosra Njah, Chuan Pham, Mohamed Cheriet, June 2020	<u>Proposed Work:</u> This paper proposes a multi-flow management scheme called Service and Resource Aware Flow Management (SRAFM) for smart digital campus networks. The scheme utilizes a fully-programmable architecture, distributed flow characterization, and centralized optimization to efficiently manage heterogeneous flows. An approximation algorithm based on Lagrangian Dual Decomposition is employed to find optimal solutions. <u>Future Work:</u> The authors aspire to augment the performance and capabilities of the proposed architecture by deploying cluster-based distributed controllers and optimizing the distributed flow characterization solution to handle vast data streams more effectively in smart networks. This will be in addition to improving the efficiency and scalability of the control plane.

[16]	William Brockelsby, Rudra Dutta, 2021	<u>Proposed Work:</u> This research paper aims to improve cybersecurity in campus networks by analysing traffic patterns and introducing architectural enhancements like strategically placed hardware or hardware-accelerated software data planes. These enhancements will regulate host-to-host communication and diminish the risk of malware propagation within network zones. <u>Future Work:</u> Future work will involve refining and optimizing of the proposed architecture for campus network cybersecurity, exploring advanced detection and mitigation techniques and adapting to emerging cyber threats. The goal is to continually enhance the security capabilities of campus networks and ensure effective protection against evolving cybersecurity risks.
[17]	Marc Hartunga, Marc Korner, 2017	<u>Proposed Work:</u> An innovative network monitoring tool is introduced for SDN substrates in the proposed work in the paper. Providing utilization charts and statistics at a flow level enhances OpenFlow controllers. The tool's capability was validated through evaluations on virtualized Mininet network as well as SDN research cluster after discussing its architecture and implementation. <u>Future Work:</u> The future work aims to further develop and enhance the SOFTmon network monitoring tool for OpenFlow networks. The focus will be on expanding its capabilities by adding support for additional network operating systems (NOS) and extending its feature set based on contributions from the community. The work will involve rigorous testing and evaluation in diverse network environments to ensure the tool's effectiveness and usability in productive SDN deployments.

III. PROPOSED SYSTEM

By utilizing cutting-edge technologies in conjunction with an intelligent orchestration framework, this proposed complex campus network management solution addresses the complexities associated with contemporary networks and provides the opportunity to improve your network performance, scalability and flexibility. This innovative method will allow campus network administrators to control and manage the campus network seamlessly due to the coordination of the various network planes.

A. Architecture

In our proposed architecture for managing a university campus-wide area network, the management plane and management interface are critical components for directing and controlling network operations. The management plane provides a central point of communication and organization for the campus network as it relates to changing device configurations, as well as serving as a communication channel between the different management interface elements and providing the means to support interoperability with future software integration.

The management plane performs an important function (ie Configuring hosts and network devices) by taking into consideration that devices come in all shapes, sizes, and capabilities (ie The Bring Your Own Device (BYOD) phenomenon). It also provides administrators with a less complicated method of deploying devices so they can manage (or "control") how the network behaves and separate users into "groups" based on what role they perform in their organisation, along with assigning them access controls and permission levels accordingly. The management plane employs strong authentication and authorisation methods to greatly reduce the risk to the network when allowing users access to the network.

The management plane is responsible for collecting and providing up-to-date information about the condition (or "status") of the network. The management plane continually monitors the network and generates alerts for "unexpected" events such as software changes, changes in configuration on network devices, attempts at unauthorised access to the network, and potential security threats to the SDN campus network.

In the data plane, the switches all support the OpenFlow protocol in full by receiving rules from the SDN controller in the control plane that defines the network policies necessary to pass packets through the network while also collecting information about the status of the network.

An automated controller translates an application's requirements into specific rules for forwarding packets in the data plane. The automated controller can create rules for the data plane through a program that follows application policies via the northbound interface.

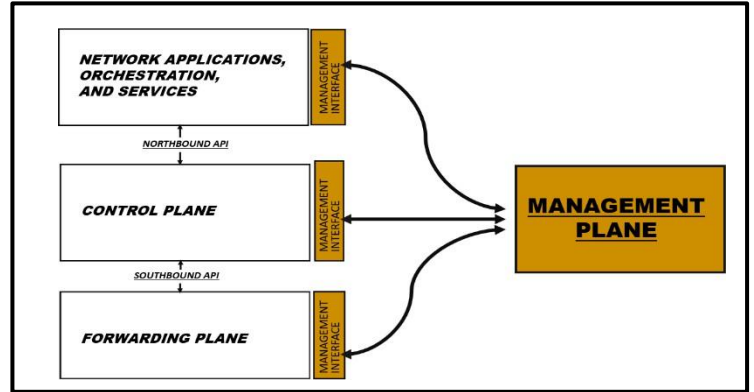


FIG. 3 High-Level Conceptual Architecture of SDN

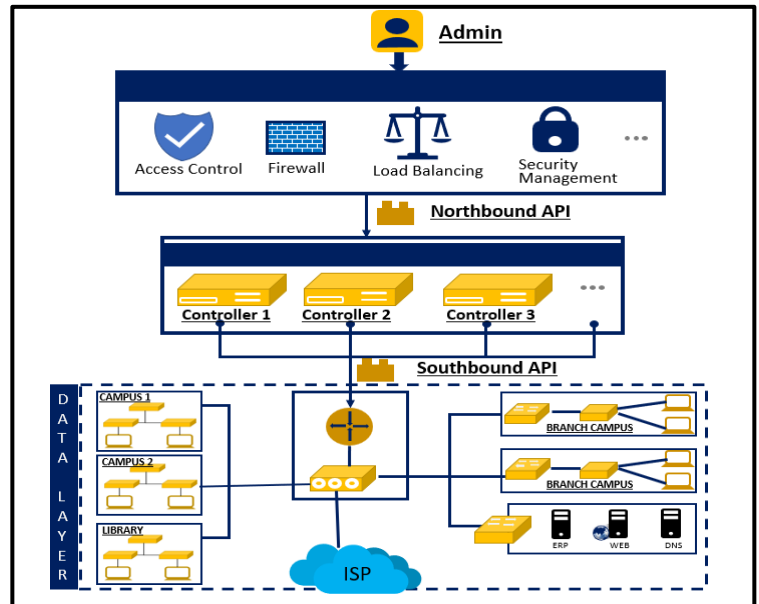


FIG. 4 Proposed SDN Framework

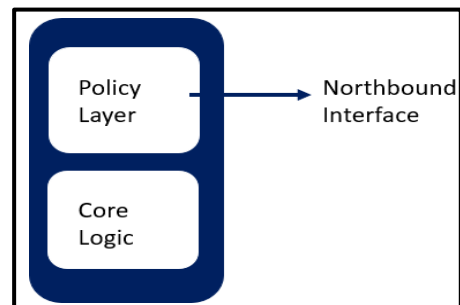


FIG. 5 Schematic Diagram of Controller

Floodlight is a common, industry proven, and high-performance Java-based OpenFlow controller that serves as your northbound interface in our architecture. It provides many benefits including excellent performance at the production level, strong industry support, and an active open-source community. Floodlight can be used by administrators to manage various OpenFlow switch parameters via their Web-based GUI or through its northbound interface API's.[18]

B. Algorithm

Our study describes a secure software-defined network (SDN) based campus network management system. In the development of this system, we have introduced three different types of algorithms: dynamic access control, anomaly detection and mitigation, and traffic engineering and quality of service (QoS) optimization. Together, these algorithms will help to improve the performance, provide increased levels of security and allow for better overall resource allocation on campus networks. Furthermore, these algorithms provide users with smooth, efficient and high-quality experiences when using the campus network.

1. DYNAMIC ACCESS CONTROL ALGORITHM:

The Dynamic Access Control Algorithm efficiently regulates network access control by evaluating both user type (e.g., faculty, student, staff) and device type being used to establish an overall access level for users on the campus. This algorithm was designed to allow for adaptive, seamless access to network resources and provide a secure environment with efficient utilization of all network resources. The Dynamic Access Control Algorithm dynamically assigns access privileges to users based on their role (e.g., faculty, student, staff) and the type of device they are using (laptop, mobile, IoT devices).

Key Components of the Dynamic Access Control Algorithm:

Input Parameters:

User_ID: A unique identifier for each user in the campus network.

User_Role: The role or category of the user, such as faculty, student, or staff.

Device_Type: The type of device the user is connecting with, like a laptop, mobile phone, or IoT device.

Access Control Policy Rules:

The algorithm defines access control policy rules based on the user roles and device types. These rules dictate the access levels and privileges users should have on the network.

Access Level Assignment:

Upon receiving the User_ID, User_Role, and Device_Type, the algorithm evaluates the corresponding access control policy rule.

Based on the rules, the algorithm assigns an appropriate access level to the user. The access levels could be "Full Access," "Limited Access," or "No Access."

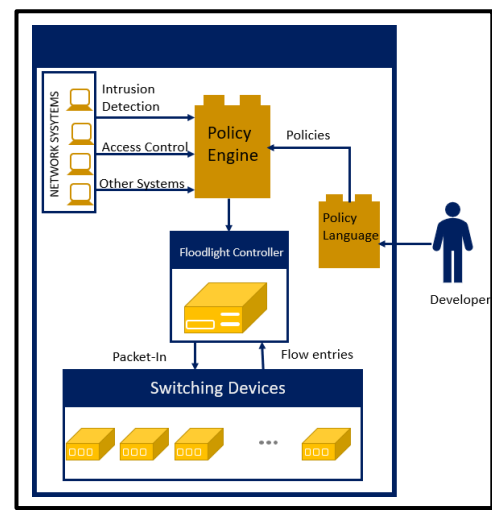


FIG. 6 Architecture of Policy Layer

Workflow of the Dynamic Access Control Algorithm:

When a user connects to the campus network, the system identifies the User_ID, User_Role, and Device_Type associated with the user.

The algorithm checks the access control policy rules to determine the appropriate access level for the user based on their User_Role and Device_Type.

If the user is identified as "Faculty," the algorithm grants them "Full Access" when using a laptop, "Limited Access" when using a mobile device, and "No Access" for other types of devices.

For "Student" users, the algorithm provides "Limited Access" on laptops and "Internet Only" access when using mobile devices, restricting access to specific resources.

For other roles not explicitly mentioned in the access control policy, the algorithm assigns "No Access," effectively denying network access.

After determining the access level, the algorithm applies the corresponding network configuration for the user, allowing or restricting their access to resources and services accordingly.

Benefits of the Dynamic Access Control Algorithm:

Efficient Resource Utilization: By dynamically assigning access levels, the algorithm optimizes resource allocation, ensuring users only have access to the resources they require based on their roles and devices.

Enhanced Network Security: The algorithm restricts unauthorized access, mitigating potential security risks and protecting sensitive data and resources within the campus network.

Adaptive Access Control: As user roles or device types change, the algorithm can adapt and adjust access levels accordingly, accommodating the dynamic nature of a campus network.

Granular Access Control: The algorithm can implement granular access control, allowing different access levels and privileges for different user roles and device types, ensuring a secure and customized network environment.

Overall, the Dynamic Access Control Algorithm plays a crucial role in managing access to network resources efficiently and securely within a campus network, making it an essential component of the proposed Secure SDN-Based Campus Network Management system.

2. ANOMOLY DETECTION AND MITIGATION ALGORITHM:

The Anomaly Detection and Mitigation Program is intended to proactively identify and mitigate anomalous/suspicious activities occurring on the campus network. The algorithm utilizes machine learning techniques and/or behavioural analysis to establish a "baseline" of normal network operations by continuously monitoring network traffic and all activities on the network. By doing so, the algorithm will detect any anomalies in network activity based on how that activity differs from the previously established baseline. When the algorithm detects an anomaly, it will initiate appropriate mitigation actions to prevent or mitigate the effect that the anomaly has on security and/or the integrity of the network.

Key Components of the Anomaly Detection and Mitigation Algorithm:

Input Parameters:

Network_Traffic: Real-time network traffic data collected from various devices and network segments.

Baseline_Network_Behavior: The historical data that represents the normal behaviour of the campus network.

Anomaly Detection:

The algorithm employs machine learning models or behavioural analysis to analyze the Network_Traffic data and identify deviations from the Baseline_Network_Behavior.

The models can include statistical methods, clustering algorithms, or more advanced machine learning techniques that have been trained to recognize patterns of normal network behaviour.

Mitigation Actions:

If an anomaly is detected, the algorithm triggers appropriate mitigation actions to prevent potential security breaches or abnormal activities.

The mitigation actions may include isolating the affected device, applying rate limiting to suspicious traffic, notifying administrators, or taking other predefined security measures.

Workflow of the Anomaly Detection and Mitigation Algorithm:

The algorithm starts by establishing

Baseline_Network_Behavior using historical network traffic data collected over a specific time period when the network is considered to be in a normal state.

As the campus network operates in real-time, the algorithm continuously collects and analyzes Network_Traffic data.

The algorithm compares the current Network_Traffic data with the Baseline_Network_Behavior to identify any significant deviations or anomalies.

If the algorithm detects an anomaly, it determines the severity of the deviation and the potential risk it poses to the network.

Based on the analysis, the algorithm triggers appropriate Mitigation_Actions to prevent or mitigate the impact of the anomaly on the network.

The algorithm may also log the detected anomaly and the corresponding mitigation actions for further analysis and auditing purposes.

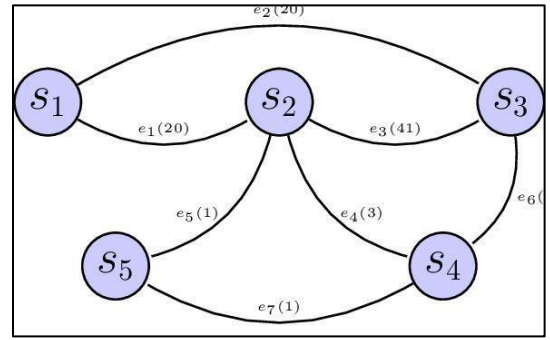


FIG. 6 Anomaly Representation

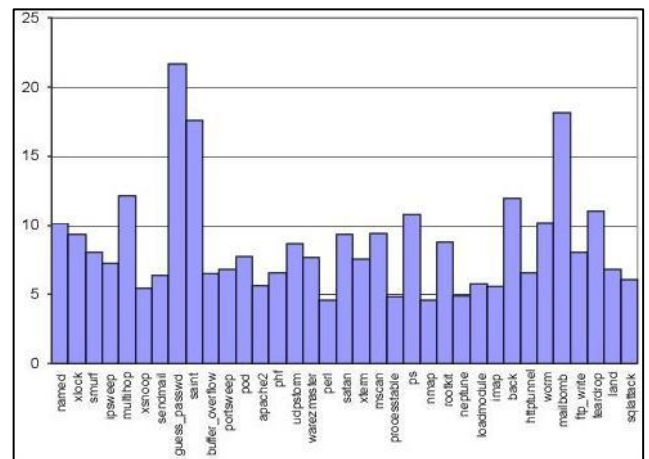


FIG. 7 Anomaly Detection Graph

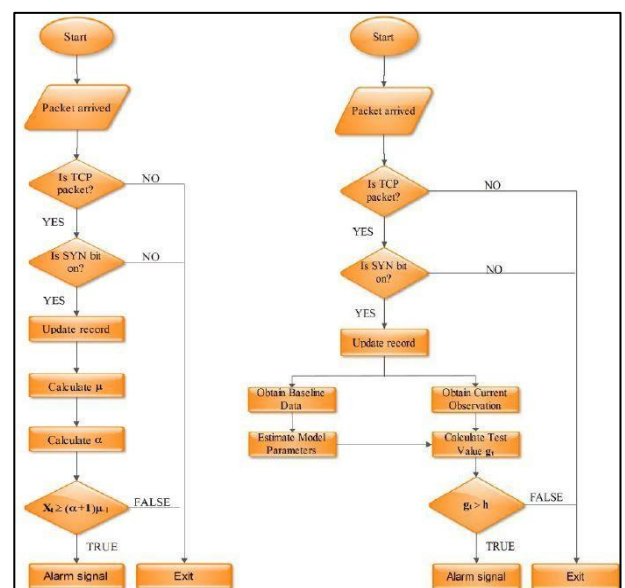


FIG. 8 Flowchart of Anomaly Detection

Benefits of the Anomaly Detection and Mitigation Algorithm:

Proactive Security: The algorithm allows network administrators to detect potential security threats before they escalate, providing an opportunity to take preventive measures.

Real-Time Response: By continuously monitoring network traffic, the algorithm can respond promptly to anomalies, minimizing the potential damage caused by security breaches.

Automation: The algorithm automates the process of anomaly detection and mitigation, reducing the need for manual intervention and enabling faster incident response times.

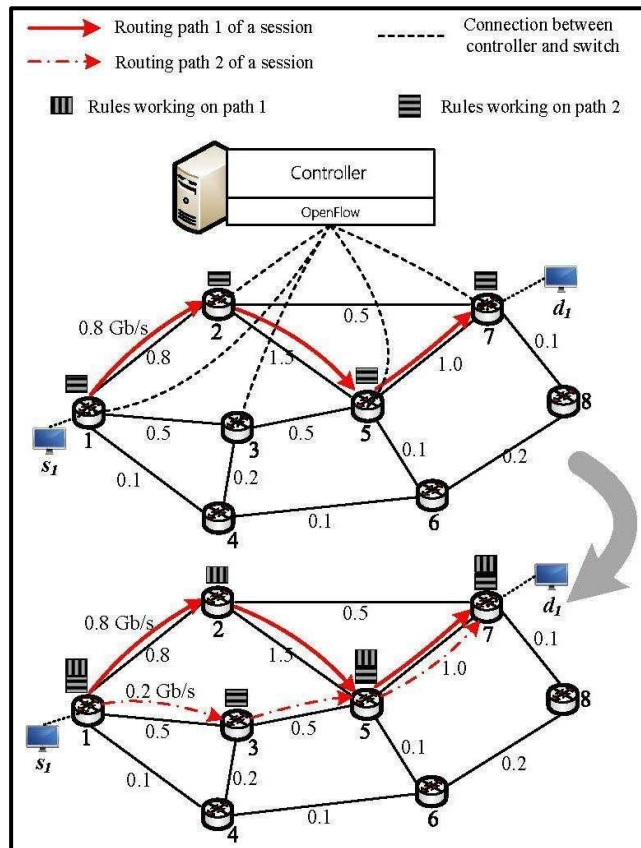
Adaptive Learning: Machine learning-based anomaly detection can adapt to changing network patterns, improving accuracy over time.

Enhanced Network Resilience: By promptly mitigating security incidents, the algorithm enhances the overall resilience of the campus network, ensuring uninterrupted services.

The Anomaly Detection and Mitigation Algorithm serve as a critical component of the proposed Secure SDN-Based Campus Network Management system, providing proactive security measures to safeguard the campus network from potential security threats and abnormal activities.

3. TRAFFIC ENGINEERING AND QoS OPTIMIZATION ALGORITHM:

Creating an overall optimized network gives you a great end-user experience and makes good use of the campus bandwidth through Traffic Engineering and QoS optimization algorithm.



Key Components of the Traffic Engineering and QoS Optimization Algorithm:

Input Parameters:

Network_Traffic: Real-time data representing the current traffic patterns within the campus network.

Application_Requirements: Specific QoS requirements and resource demands of various applications and services running on the network.

QoS_Policies: Predefined QoS policies that prioritize certain types of traffic based on their importance.

Traffic Engineering:

An algorithm analyzes Network_Traffic data to find congestion, bottlenecks, or possible performance problems in a network.

After finding these problems, the algorithm will change how data travels over the network and traffic flows to improve how data is delivered, increasing the efficiency of available resources and the performance of the entire network.

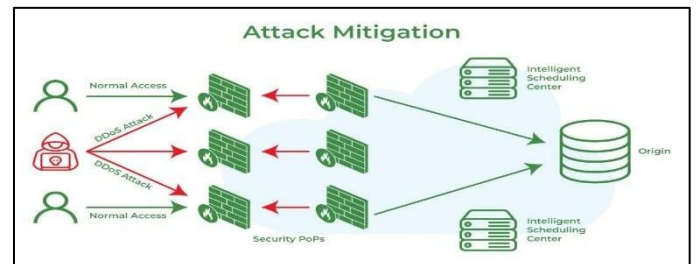


FIG. 9 Attack Mitigation

QoS Policy Enforcement:

The algorithm applies the relevant QoS policies to give higher priority to critical traffic for all applications or services found in the Application_Requirements.

QoS policies can include, but are not limited to, bandwidth reservation, rate limiting, packet prioritization, and a variety of other methods used to satisfy an application's specific QoS needs.

Workflow of the Traffic Engineering and QoS Optimization Algorithm:

The algorithm starts by evaluating the current traffic data of the network to see where there may be congestion and/or possible performance issues in the network itself. Once evaluated, the algorithm will adjust the way traffic flows through the various paths of the network dynamically, in order to create an even distribution of traffic while avoiding paths that could become congested and causing bottlenecks.

For each application or service that is defined in Application_Requirements, there are also corresponding QoS_Policies that the algorithm uses to guarantee that critical traffic will receive the appropriate network resource(s).

The algorithm will regularly check the network to look for any alterations in traffic patterns or demands from the applications running on it. If an application is found that has a unique quality of service to achieve then it can change how the paths and traffic flow through them based on what has been added.

Benefits of the Traffic Engineering and QoS Optimization Algorithm:

Optimized Network Performance: The algorithm ensures that network traffic is efficiently distributed, minimizing congestion and reducing packet loss, leading to improved overall network performance.

Seamless User Experience: By prioritizing critical applications and services, the algorithm guarantees that users experience reliable and uninterrupted access to essential resources.

Efficient Resource Utilization: The algorithm optimizes

bandwidth allocation, ensuring that network resources are used effectively and wastage is minimized.

Dynamic Adaptation: The algorithm can adapt to changes in network traffic and application requirements, making it suitable for managing dynamic campus networks with varying user demands.

The Traffic Engineering and QoS Optimization Algorithm has an important role in the Secure SDN-Based Campus Network Management system to help improve the entire network experience, improve the overall performance of both Users and Resource Allocation, while meeting QoS metrics required for each application or service being used.

In conclusion, since the "most efficient" algorithm will differ based on the priorities and goals of the Campus Network, if Security is a major priority, then the Anomaly Detection and Mitigation Algorithm would likely be the most efficient option. If Resource Optimization and User Experience are of high importance, then it would be the Traffic Engineering and QoS Optimization Algorithm that is most efficient. If there is a need to frequently change Roles or to use Adaptive Access Control, then the Dynamic Access Control Algorithm would have a high level of usefulness.

In a good situation, a set of combined algorithms working together in a well-organized secure SDN-based campus network management system should provide optimal efficiency, since they will each handle a specific area of concerns with network management and security. Each algorithm's degree of effectiveness will depend on how well they have been implemented and integrated into the overall network topology. Also, to maintain the efficiency and effectiveness of a logic circuit over time, regular monitoring and fine-tuning are necessary.

In this section we will discuss the results and accomplishments of our research study that focuses on how to manage complicated campus networks through the implementation of a Secure Software Defined Networking (SDN) system. The proposed solution will overcome the complex issues faced by today's networking environments by providing cutting edge technology and an intelligent orchestration framework. The system will improve operation of the network as well as provide greater security, thus providing network efficiency, scalability and flexibility.

1. PROPOSED SYSTEM ARCHITECTURE

The Network Management System we propose is a fully integrated system, with a complete architecture for managing and controlling university-wide/county-wide campus networks. The architecture has three distinct planes (management, service, and control planes) that all perform important functions to ensure coordinated and effective management of the network for the campus/university.

The Management Plane is the central hub of the campus/university network management system that organizes and oversees the operations of the network. It allows changes to be made to some of the network device settings, facilitates the flow of information between the different network planes, and provides compatibility for integrating new software and tools. The Management Plane also provides network devices and hosts the ability to be automatically configured, taking into account device type and BYOD, making it much easier to deploy devices in the network as well as enforce access control policies. All of these features increase the security of the network by providing strong authentication and authorization processes.

The data plane leverages OpenFlow protocol-compliant switching devices to implement network policies, ensuring smooth packet forwarding and continuous collection of network status information. Additionally, the control plane is responsible for translating application requirements into forwarding rules, which are executed on the data plane. This automated operation is supported by a programming language or protocol available through the northbound interface, ensuring compatibility with SDN-compliant network devices via the OpenFlow protocol.

2. ALGORITHMIC CONTRIBUTIONS

In this study we created three main algorithms that work together to improve network management, network security, and the ability to provide quality service to users of the campus network.

The Dynamic Access Control Algorithm (DACA) provides network administrators with a way to efficiently manage all users who connect to the network by dynamically providing user-specific access levels and privileges depending on the user's role (faculty, student, or staff) as well as the type of device being used (laptop, mobile, or IoT). As a result, the DACA algorithm will allow for a more efficient use of available resources within the network, while at the same time increasing the security of the network and adapting as the user roles or device types change.

3. IMPLICATIONS AND SIGNIFICANCE

Our research yields evidence to support both the feasibility and applicability of the Secure SDN-based Campus Network Management System. The system has been implemented as described in both architecture and algorithms with regard to network resource management (efficiency), distribution of traffic (load balancing), and reduction of congestion (by means of prevention). This implementation has resulted in:

Optimized Network Performance: This architecture and employed algorithms optimize overall network performance for efficient use of available bandwidth, by better balancing the load across all network resources thereby avoiding (or at least reducing) congestion on said resources. *Enhanced Security:* The proposed Dynamic Access Control Algorithm and Anomaly Detection and Mitigation Algorithm bolster network security by dynamically managing access control and proactively identifying and mitigating potential security threats.

The Traffic Engineering and QoS Optimization Algorithm creates a customized user experience by giving priority to essential apps and enforcing QoS policies specific to the needs of the app.

Adaptability: The system's ability to adapt to changing network conditions, user roles, and types of devices makes it an effective management solution for the dynamic campus network.

REFERENCES

1. Zhao Zhang, Hailong Li, Siqi Dong & Lei Hu (2018). Software Defined Networking (SDN) Research Review. 10.2991/mecae-18.2018.129.
2. Rajat Chaudhary, Anish Jindal, Gagangeet Singh Aujla, Shubhani Aggarwal, Neeraj Kumar, and Kim-Kwang Raymond Choo. 2019. BEST: Blockchain-based secure energy trading in SDN-enabled intelligent transportation system. *Comput. Secur.* 85, C (Aug 2019), 288–299. <https://doi.org/10.1016/j.cose.2019.05.006>
3. Pradeep Sharma, & Dr. S.S. Tyagi, (2019). Improving Security through Software Defined Networking (SDN): AN SDN based Model. *International Journal of Recent Technology and Engineering (IJRTE)*. 8. 295-300. 10.35940/ijrte.D6814.118419.
4. Piyush Kumar Sharma, Devashish Gosain, Himanshu Sagar, Chaitanya Kumar, Aneesh Dogra, Vinayak Naik, H.B. Acharya, Sambuddho Chakravarty. "SiegeBreaker: An SDN Based Practical Decoy Routing System." *Proceedings on Privacy Enhancing Technologies* 2020.3 (2020) 243-263
5. Faridullah Amarkhil, Prashansa Taneja, 2020, A Research Paper of Security Enforcement Policy for (SDN) (WLAN) Software Defined Network, *INTERNATIONAL JOURNAL OF ENGINEERING RESEARCH & TECHNOLOGY (IJERT)* Volume 09, Issue 06 (June 2020)
6. Khamaiseh, Samer, Abdullah Al-Alaj, Mohammad Adnan, and Hakam W. Alomari. 2022. "The Robustness of Detecting Known and Unknown DDoS Saturation Attacks in SDN via the Integration of Supervised and Semi-Supervised Classifiers" *Future Internet* 14, no. 6: 164. <https://doi.org/10.3390/fi14060164>
7. J. Zhang, M. Ye, Z. Guo, C. -Y. Yen and H. J. Chao, "CFR-RL: Traffic Engineering With Reinforcement Learning in SDN," in *IEEE Journal on Selected Areas in Communications*, vol. 38, no. 10, pp. 2249-2259, Oct. 2020, DOI: 10.1109/JSAC.2020.3000371.
8. Hnin Thiri Zaw, & Aung Htein Maw. (2019). Traffic management with elephant flow detection in software defined networks (SDN). *International Journal of Electrical and Computer Engineering (IJECE)*, 9(4), 3203–3211. <https://doi.org/10.11591/ijece.v9i4.pp3203-3211>
9. Md. Jahidul Islam, Anichur Rahman, Sumaiya Kabir, Md. Razaul Karim, Uzzal Kumar Acharjee, Mostofa Kamal Nasir, Shahab S. Band, Mehdi Sookhak & Shaoen Wu. (2021). Blockchain-SDN-Based Energy-Aware and Distributed Secure Architecture for IoT in Smart Cities. *IEEE Internet of Things Journal*. 9. 3850 - 3864. 10.1109/JIOT.2021.3100797.
10. Ashutosh Kumar Singh, Naveen Kumar, Shashank Srivastava, "PSO and TLBO based Reliable Placement of Controllers in SDN", *International Journal of Computer Network and Information Security (IJCNIS)*, Vol.11, No.2, pp.36-42, 2019. DOI: 10.5815/ijcnis.2019.02.05
11. Tiago Linhares, Ahmed Patel, Ana Luiza Barros and Marcial Fernandez, "SDN Truth: Innovative DDoS Detection Scheme for Software-Defined Networks (SDN)", DOI: <https://doi.org/10.21203/rs.3.rs-2223091/v1>
12. A. Al-Jawad, I. -S. Comşa, P. Shah, O. Gemikonakli and R. Trestian, "An Innovative Reinforcement Learning-Based Framework for Quality-of-Service Provisioning Over Multimedia-Based SDN Environments," in *IEEE Transactions on Broadcasting*, vol. 67, no. 4, pp. 851-867, Dec. 2021, DOI: 10.1109/TBC.2021.3099728.
13. Y. Njah, C. Pham and M. Cheriet, "Service and Resource Aware Flow Management Scheme for an SDN-Based Smart Digital Campus Environment," in *IEEE Access*, vol. 8, pp. 119635-119653, 2020, DOI: 10.1109/ACCESS.2020.3005569.
14. W. Brockelsby and R. Dutta, "Traffic Analysis in Support of Hybrid SDN Campus Architectures for Enhanced Cybersecurity," 2021 24th Conference on Innovation in Clouds, Internet and Networks and Workshops (ICIN), Paris, France, 2021, pp. 41-48, DOI: 10.1109/ICIN51074.2021.9385530.
15. Marc Hartung, & Marc Körner. (2017). SOFTmon - Traffic Monitoring for SDN. *Procedia Computer Science*. 110. 516-523. 10.1016/j.procs.2017.06.138.
16. Udo, Edward & Isong, Etebong & Nyoho, Emmanuel. (2020). Software Defined Networking Framework for Campus Network Management.